looking for all the points within a hypersphere.

Listing 12.6 DBSCAN Clustering

```
function dbscan(points, eps, minPoints)
                  #1
  currentIndex ← 0
                  #2
  clusterIndices ← 0 (" p ∈
points)                                          #3
  kd ← new KdTree(points)
                                                    #4
  for p in points do
                  #5
    if clusterIndices != 0
then                                               #6
      continue
    toProcess ← {p}
                          #7
    clusterIndices[p] ← -1
                                                    #8
    currentIndex ← currentIndex +
1                                              #9
    for q in toProcess do
                  #10
      neighbors ← kd. pointsInSphere(q,
eps)                                       #11
      if |neighbors| < minPoints then
                  #12
        continue
      clusterIndices[q] ←
currentIndex                                        #13
      toProcess ← toProcess + {w in neighbors |
clusterIndices[w] ≤ 0}        #14
  return clusterIndices
                                             #15
```